Bihar Engineering University, Patna

B.Tech 8th Semester Examination, 2024(S)

Course: B.Tech
Code: 103404

Subject: Signals and Systems

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Answer the following question (any seven only):-*** **[2 x 7 = 14]**

(a) What is the significance of periodicity in signal properties?

(b) Define the system property of causality in terms of input-output behavior.

(c) How does convolution apply to continuous and discrete-time LTI systems?

(d) What is the difference between continuous and discrete-time signals?

(e) What is the role of the state transition matrix in state-space analysis?

(f) How does the Fourier series represent periodic signals?

(g) What is the relationship between the Fourier Transform and convolution in the frequency domain?

(h) What does the Laplace transform reveal about the system's behavior and stability?

(i) How does aliasing occur during the sampling process, and what are its effects?

(j) What is the significance of the Sampling Theorem in signal processing?

Q.2 Explain the different properties of signals such as periodicity, absolute integrability, determinism, and stochastic character and provide examples for each. How do these properties impact the analysis of signals in various applications? **[14]**

Q.3 Describe the various system properties, including linearity (additivity and homogeneity), shift-invariance, causality, stability and realizability. How do these properties influence the behavior and design of systems in signal processing? **[14]**

Q.4 Discuss the behavior of continuous and discrete-time LTI systems. How are impulse response and step response used to characterize these systems and what role does convolution play in understanding input-output behavior for aperiodic inputs? **[14]**

Q.5 Define the concept of state-space representation in systems theory. How is a system represented through differential and difference equations and what advantages does the state-space representation offer for analyzing multi-input, multi-output systems? **[14]**

Q.6 Discuss the Sampling Theorem and its implications for signal processing. What are the effects of aliasing and how does it affect the reconstruction of signals? Explain the differences between ideal interpolation, zero-order hold and first-order hold in the reconstruction process. **[14]**

Q.7 Explain the concept of the Laplace Transform in the analysis of continuous-time signals and systems. How are system functions, poles and zeros used to analyze system stability and behavior in the Laplace domain? **[14]**

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Q.8 Describe the relationship between the Fourier Transform and convolution. How does the Fourier Transform help in analyzing the magnitude and phase response of a system and what is meant by Fourier domain duality? **[14]**

Q.9 Write Short notes on ***any two*** of the following: **[7 x 2 = 14]**

(a) Significance of the unit impulse signal in the analysis of linear time-invariant systems

(b) Concept of shift-invariance apply to both continuous and discrete-time systems

(c) Role of poles and zeros in the Laplace and z-Transforms for system analysis?

(d) Discrete Fourier Transform (DFT)

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